

Calc 1 – The Integral

Worksheet A

1. Compute $\int (3x^2 - 4x + 1) dx$.
2. Compute $\int_0^1 (e^x + 1) dx$.
3. Compute $\int \left(\frac{1}{x} + \sin x\right) dx$.
4. Find $F'(x)$ where $F(x) = \int_0^{x^2} \cos(t) dt$.
5. Find the average value of $f(x) = x^2$ on $[0, 3]$.

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Worksheet B

1. Compute $\int_0^4 \sqrt{x} \, dx$.
2. Compute $\int_{-1}^1 (x^3 + x) \, dx$.
3. Compute $\int \sec^2 x \, dx$.
4. Find $F'(x)$ where $F(x) = \int_x^{x^2} e^{-t^2} \, dt$.
5. Approximate $\int_0^4 (x^2 + 1) \, dx$ using the right Riemann sum with $n = 4$.

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Answer Key (Worksheets A & B)

Worksheet A.

1. $x^3 - 2x^2 + x + C$.
2. $[e^x + x]_0^1 = (e + 1) - 1 = e$.
3. $\ln|x| - \cos x + C$.
4. Leibniz: $F'(x) = \cos(x^2) \cdot 2x = 2x \cos(x^2)$.
5. $\bar{f} = \frac{1}{3} \int_0^3 x^2 dx = \frac{1}{3} \cdot 9 = 3$.

Worksheet B.

1. $[\frac{2}{3}x^{3/2}]_0^4 = \frac{2}{3} \cdot 8 = 16/3$.
2. Odd integrand on symmetric interval: 0 (or compute $[\frac{x^4}{4} + \frac{x^2}{2}]_{-1}^1 = 0$).
3. $\tan x + C$.
4. Leibniz: $F'(x) = e^{-x^4} \cdot 2x - e^{-x^2} \cdot 1 = 2xe^{-x^4} - e^{-x^2}$.
5. $\Delta x = 1$, sample at $x_i = 1, 2, 3, 4$. Sum = $1(2 + 5 + 10 + 17) = 34$ (exact value = $4^3/3 + 4 = 76/3 \approx 25.33$).